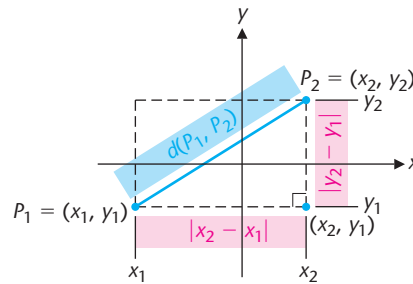


► **Figure 2** Distance between two points.



► **THEOREM 1** Distance Formula

The distance between $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

EXAMPLE

2

Using the Distance Formula

Find the distance between the points $(-3, 5)$ and $(-2, -8)$.*

SOLUTION

Let $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (-2, -8)$. Then,

$$\begin{aligned} d &= \sqrt{[(-2) - (-3)]^2 + [(-8) - 5]^2} \\ &= \sqrt{(-2 + 3)^2 + (-8 - 5)^2} = \sqrt{1^2 + (-13)^2} = \sqrt{1 + 169} = \sqrt{170} \end{aligned}$$

Notice that if we choose $(x_1, y_1) = (-2, -8)$ and $(x_2, y_2) = (-3, 5)$, then

$$d = \sqrt{[(-3) - (-2)]^2 + [5 - (-8)]^2} = \sqrt{1 + 169} = \sqrt{170}$$

so it doesn't matter which point we designate as P_1 or P_2 .

MATCHED PROBLEM 2

Find the distance between the points $(6, -3)$ and $(-7, -5)$.

► **Midpoint of a Line Segment**

The **midpoint** of a line segment is the point that is equidistant from each of the endpoints. A formula for finding the midpoint is given in Theorem 2. The proof is discussed in the exercises.

► **THEOREM 2** Midpoint Formula

The midpoint of the line segment joining $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

The point M is the unique point satisfying

$$d(P_1, M) = d(M, P_2) = \frac{1}{2}d(P_1, P_2)$$

*We often speak of the point (a, b) when we are referring to the point with coordinates (a, b) . This shorthand, though not technically accurate, causes little trouble, and we will continue the practice.